CAN Protocol

CAN Error Counters

- TEC -- Ack Error, Bit Error
- REC -- Stuffing Error, Frame Error, CRC Error

CAN Bus States

- Error Active: 0 < TEC and REC <= 127
  - Normal communication
- Error Passive: 128 <= TEC or REC <= 255
  - Limited communication
  - Error reporting is not actively done
- Bus Off: TEC > 255
  - Stop all transmissions and receptions

Recovering from Bus Off state

- Detection of Bus off state
  - TEC > 255
  - CAN_BusOff_Handler()
- Wait for fixed number of Bus Idle state
  - Bus Idle state = 11 recessive bits consecutive
  - Wait for 128 such idle periods
- Reset Error counters
  - TEC = 0
  - REC = 0
- Reinitialize CAN controller
  - Disable and re-enable CAN controller
  - Reconfig CAN controller (initialization)

- Join CAN network
  - Start normal communication (Error active state)

Terminal resistors on CAN bus

- 120 ohm resistors
- Impedence match
- Signal reflection (echo)
- Signal integrity

Big Banging

- Chaning GPIO bits periodically as per requirement -- Bit Banging.
- Typically done to implement communication Protocols in software.
  - Use timer and change GPIO pin to 1/0 as per Protocol timing and requirement.
- Possible in every micro-controller.

Bit Banding

- Each bit on memory location/IO register is mapped to one word of address space.
- Any write on that word, will reflect on correspinding bit and reverse (read).
- Feature of Cortex-M arch.
  - RAM bit band region (1 MB) --> RAM bit band alias region (32 MB)
  - Peri bit band region (1 MB) --> Peri bit band alias region (32 MB)
- Atomic update of IO register.
  - regr |= (1 << 4)
    - load regr
    - bitwise or
    - update regr
  - bit_alias_word = 1
    - read from bus, and change and write on bus -- in same bus cycle.
- We can use bit banding to periodically update some bit -- bit banging.